

MOHAMED BASSAM

AI Engineer | Agentic AI | Machine Learning | Computer Vision | NLP

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Portfolio | GitHub

PROFESSIONAL SUMMARY

AI Engineer and Full-Stack Developer with hands-on experience building and deploying end-to-end intelligent systems across agentic AI, NLP, computer vision, and secure SaaS architecture. Skilled in taking solutions from data preparation and model training with PyTorch, TensorFlow, and YOLO11 through production APIs and interactive applications using FastAPI, Flask, and Next.js. Creator of Dark Agent, a governed multi-agent mission-control platform with adversarial validation, tenant-isolated retrieval, and live telemetry.

TECHNICAL SKILLS

Programming & DB:

Python TypeScript / JavaScript SQL SQLite PostgreSQL

Agentic AI & LLMs:

LangGraph Multi-Agent Orchestration Hybrid RAG MCP OpenAI API Anthropic API

Machine Learning & NLP:

TensorFlow / Keras PyTorch Scikit-learn XGBoost LightGBM LSTM BERT
Hugging Face

Computer Vision:

YOLO11 OpenCV CUDA

Backend & SaaS:

FastAPI Flask REST APIs WebSockets Pydantic Clerk Stripe APScheduler

Retrieval & Data:

pgvector BM25 / RRF Pandas NumPy

Frontend, Cloud & DevOps:

React Next.js Tailwind CSS Docker Azure Git / GitHub Pytest n8n

SELECTED AI PROJECTS

FLAGSHIP SYSTEM | GOVERNED AGENTIC AI

Dark Agent - Governed Multi-Agent AI Mission Control Platform

Agentic AI

Tech: Python, FastAPI, LangGraph, Next.js, TypeScript, PostgreSQL / pgvector, BM25 / RRF, WebSockets, Clerk, Stripe, Docker, Azure, MCP, OpenAI, Anthropic

- Architected a governed LangGraph workflow with a Planner, parallel specialists, Synthesizer, and adversarial Red Team validator, plus WebSocket streaming and human plan approval.
- Implemented tenant-isolated hybrid retrieval using pgvector semantic search, BM25 keyword ranking, and Reciprocal Rank Fusion, with episodic memory distillation and client-side PII sanitization.
- Secured the SaaS control plane with Clerk authentication, idempotent Stripe webhooks, database-enforced model routing, rate limits, deterministic audit logs, and zero-retention Ghost Mode.
- Built Docker and Azure production foundations with Service Bus workers, Key Vault-backed read-only MCP connectors, provider fallback, 309 backend tests, and a clean Next.js production build.

SELECTED AI PROJECTS - CONTINUED

AI Stock Price Prediction - AI Stock Market Analysis & Paper Trading Platform

Graduation Project

Tech: Python, Flask, TensorFlow / Keras, LSTM, Scikit-learn, XGBoost, LightGBM, Pandas, NumPy, REST APIs, Alpaca API, Chart.js, Time-Series Forecasting

- Built an AI-powered stock analysis and paper-trading platform using an ensemble of LSTM, XGBoost, LightGBM, and ExtraTrees models to forecast market trends.
- Integrated Alpaca Paper Trading for simulated execution, live portfolio tracking, trade history, duplicate-order prevention, and protected take-profit / stop-loss exits.
- Engineered market features from RSI, EMA, SMA, MACD, ROC, Bollinger Bands, ATR, VWAP, sentiment logic, and volatility analysis.
- Created a quantitative risk engine with technical indicators, historical backtesting, walk-forward validation, confidence scoring, and actionable BUY / SELL / HOLD setups.
- Designed an explainable Chart.js dashboard showing prediction reasoning, price trajectories, entry, stop-loss, take-profit, risk level, and real-time P/L.

AI Industrial Safety Monitoring System

Computer Vision

Tech: Python, YOLO11, OpenCV, n8n, Webhooks, CUDA

- Engineered a real-time industrial safety monitoring system using YOLO11 and OpenCV.
- Implemented PPE workflows for Hardhat / No-Hardhat verification in workplace environments.
- Programmed Safe Zone and Danger Zone spatial rules with automated violation logs.
- Added dual-model validation to improve detection reliability and reduce false alerts.
- Connected n8n workflows to trigger Gmail incident alerts with timestamped evidence images.
- Optimized real-time inference performance with Torch CUDA GPU acceleration.

AI Study Assistant - NLP Web Application

NLP

Tech: Python, Flask, Hugging Face Transformers, BART, PDF Processing

- Developed an NLP summarization application using the pre-trained Hugging Face BART model.
- Built Flask endpoints for direct text input and multi-page PDF document processing.
- Designed chunking and multi-pass summarization logic to process long documents reliably within model memory limits.

AI Writer Web Application

Full-Stack AI

Tech: Python, Flask, SQLite, REST APIs, PythonAnywhere

- Developed and deployed a full-stack AI SaaS application for content generation, rewriting, and text improvement.
- Implemented secure user authentication, session management, and persistent generation history with Flask and SQLite.
- Developed REST endpoints for core generation logic and asynchronous user interactions.
- Deployed the live production application successfully on PythonAnywhere.

PROFESSIONAL EXPERIENCE

Machine Learning Intern

2024

CodeAlpha

- Developed, tested, and evaluated machine-learning models using Support Vector Machines and neural networks.
- Applied TensorFlow and OpenCV to practical object-recognition and data-classification tasks.
- Evaluated performance with ROC curves, confusion matrices, precision, and recall.
- Used NumPy, Matplotlib, Pandas, and Keras to process datasets and visualize experimental results.

EDUCATION

Bachelor of Computer Science (Artificial Intelligence)

Graduated: July 2025

Arab Open University (AOU) & The Open University (UK)

Dual Degree Program verified by The Open University, UK.